

MARL-Based Supervisory Reactive Power Control for Post-Fault Voltage Recovery in Active Distribution Networks

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Abstract—This paper proposes a Multi-Agent Reinforcement Learning (MARL)-based supervisory controller for post-fault voltage recovery in Active Distribution Networks (ADNs) with photovoltaic (PV) inverter resources. The controller coordinates inverter reactive power setpoints after fault clearing, while its actions are held fixed during the fault-on interval to avoid interference with fast inverter inner-loop and fault ride-through dynamics. The approach is evaluated using RMS dynamic simulations on a modified IEEE 33-bus distribution network over 101 operating and fault scenarios. Compared with droop and fixed-parameter PI baselines, the MARL policies mainly improve transient recovery behavior, particularly in terms of settling time, post-fault time within the admissible voltage band, and voltage violation area. The fixed-PI controller provides the best final voltage accuracy and shortest first recovery time, but exhibits longer settling behavior. Among the learned policies, Fast MARL achieves the strongest recovery-oriented performance, whereas Balanced MARL provides a more conservative response with lower voltage peaks and reduced peak reactive power injection. The results indicate that MARL-based supervisory coordination can provide flexible trade-offs between recovery quality and control effort.

Index Terms—Active Distribution Networks (ADNs), Multi-Agent Reinforcement Learning (MARL), post-fault voltage recovery, PV inverters, reactive power control, RMS simulation, voltage regulation.

I. INTRODUCTION

The increasing penetration of inverter-based Distributed Energy Resources (DERs) is changing the dynamic behavior of Active Distribution Networks (ADNs) [1]. In feeders with high photovoltaic (PV) generation, voltage profiles become more sensitive to active and reactive power variations, especially during rapid operating-point changes and after disturbances [2]. Following a fault and its clearance, the voltage trajectory may exhibit delayed recovery, overshoot, or prolonged operation outside acceptable limits [3]. These effects are influenced by inverter current limits, reactive power capability, and the location of DERs [4].

Conventional voltage regulation schemes, such as local Volt-VAR droop control, fixed-parameter proportional-integral

controllers, and On Load Tap Changers (OLTCs), remain widely used because of their simplicity and reliability [5]. However, their performance depends strongly on the operating point, controller tuning, disturbance location, and network topology [6]. A controller tuned for one condition may become slow, conservative, or poorly coordinated under another [7]. This limitation becomes more important when several inverter-based resources are distributed across the feeder, since local actions can interact through the network and may not provide the best system-level post-fault recovery [8].

Recent studies have addressed post-fault voltage behavior in converter-dominated systems, including temporary overvoltage in VSC-rich networks [9], data-driven post-fault voltage-stability prediction in hybrid AC/DC microgrids [10], and graph-based Multi-Agent Reinforcement Learning (MARL) for transmission-level emergency voltage control [11]. In parallel, MARL methods have demonstrated steady-state or quasi-static Volt-VAR control on feeders up to 141 buses [12]–[14]. However, distribution-level post-fault supervisory coordination of PV inverter reactive power, with explicit separation between fault-on FRT dynamics and post-fault recovery control, remains less explored.

MARL is a suitable framework for this problem because distributed PV inverters can be modeled as agents that coordinate reactive power support from local observations. However, post-fault voltage recovery differs from conventional steady-state Volt-VAR control because the supervisory controller must act only after fault clearing and must not interfere with fast fault ride-through or converter inner-loop dynamics. Therefore, this paper investigates a MARL-based supervisory control strategy that coordinates PV inverter reactive power setpoints after fault clearance at a slower supervisory timescale. The approach is evaluated on a modified IEEE 33-bus distribution network using RMS dynamic simulations and compared with droop and fixed-PI transient voltage-support baselines. The focus is on establishing the post-fault supervisory control framework, while scalability to larger feeders is left for future work. The main contributions of this paper are summarized as follows:

- A post-fault supervisory control framework is formulated for inverter-based voltage recovery in ADNs, explicitly separating the fast fault-on inverter response from the slower post-fault recovery control.

This project has received funding from the European Union's Horizon Europe Framework Programme (HORIZON) under the GA n. 101120278 - DENSE.

- A MARL-based controller is developed to coordinate distributed PV inverter reactive power support after fault clearing, with two reward settings: a recovery-oriented Fast RL policy and a more conservative Balanced RL policy.
- The proposed controller is evaluated against droop and fixed-PI transient voltage-support baselines using RMS dynamic simulations over 101 operating and fault scenarios on a modified IEEE 33-bus distribution network.

The remainder of the paper is organized as follows. Section II formulates the problem. Section III presents the proposed control framework. Section IV provides the numerical results, and Section V concludes the paper.

II. PROBLEM FORMULATION

The objective is to coordinate inverter-based reactive power support after fault clearing so that feeder voltages return to the admissible operating range while avoiding excessive reactive power usage. The formulation below defines this post-fault recovery problem independently of the specific control method. This work considers an ADN with distributed PV. The distribution feeder is represented by a graph

$$\mathcal{G} = (\mathcal{N}, \mathcal{E}), \quad (1)$$

where $\mathcal{N}$ is the set of buses and $\mathcal{E}$ is the set of distribution lines. Let $N = |\mathcal{N}|$ denote the number of buses. The voltage magnitude at bus $i \in \mathcal{N}$ and time t is denoted by $v_i(t)$.

The network contains a set of controllable PV inverters denoted by $\mathcal{C} \subseteq \mathcal{N}$. Each controllable inverter can provide reactive power support subject to its apparent power rating. For inverter $k \in \mathcal{C}$, the active power generation at time t is denoted by $p_k^{\text{PV}}(t)$, and the apparent power rating is denoted by S_k^{max} . The available reactive power capability is then given by

$$q_k^{\text{max}}(t) = \sqrt{(S_k^{\text{max}})^2 - (p_k^{\text{PV}}(t))^2}. \quad (2)$$

The reactive power injection $q_k(t)$ must satisfy

$$-q_k^{\text{max}}(t) \leq q_k(t) \leq q_k^{\text{max}}(t), \quad \forall k \in \mathcal{C}. \quad (3)$$

A disturbance is assumed to occur at time t_f and to be cleared at time t_c . The post-fault recovery interval is defined as

$$\mathcal{T}_{\text{pf}} = \{t \mid t \geq t_c, t \leq t_c + T_{\text{rec}}\}, \quad (4)$$

where T_{rec} is the considered recovery horizon. The objective of the controller is to improve the voltage trajectory during this post-fault interval while respecting inverter capability limits.

The admissible voltage operating range is defined by lower and upper voltage limits:

$$v^{\min} \leq v_i(t) \leq v^{\max}, \quad \forall i \in \mathcal{N}. \quad (5)$$

For each bus, the voltage violation magnitude at time t is defined as

$$\Delta v_i(t) = \max(0, v^{\min} - v_i(t)) + \max(0, v_i(t) - v^{\max}). \quad (6)$$

The total voltage violation over the feeder is then

$$J_{\text{vio}}(t) = \sum_{i \in \mathcal{N}} \Delta v_i(t). \quad (7)$$

In addition to keeping voltages within the admissible band, the controller should reduce the deviation from the nominal voltage. The average voltage tracking error is defined as

$$J_{\text{err}}(t) = \frac{1}{N} \sum_{i \in \mathcal{N}} |v_i(t) - v^{\text{ref}}|, \quad (8)$$

where v^{ref} is the nominal voltage reference, typically set to 1.0 pu.

The use of reactive power support is also included in the formulation to avoid unnecessarily aggressive control actions. The reactive power effort is expressed as

$$J_q(t) = \sum_{k \in \mathcal{C}} |q_k(t)|. \quad (9)$$

The general post-fault voltage recovery problem can therefore be written as

$$\begin{aligned} \min_{\{q_k(t)\}} \quad & \sum_{t \in \mathcal{T}_{\text{pf}}} (\lambda_{\text{vio}} J_{\text{vio}}(t) + \lambda_{\text{err}} J_{\text{err}}(t) + \lambda_q J_q(t)) \\ \text{s.t.} \quad & -q_k^{\text{max}}(t) \leq q_k(t) \leq q_k^{\text{max}}(t), \quad \forall k \in \mathcal{C}, t \in \mathcal{T}_{\text{pf}}, \\ & v_i(t) = f_i(\mathbf{p}(t), \mathbf{q}(t), \mathcal{G}, \boldsymbol{\xi}(t)), \quad \forall i \in \mathcal{N}, t \in \mathcal{T}_{\text{pf}}. \end{aligned} \quad (10)$$

Here, λ_{vio} , λ_{err} , and λ_q are weighting coefficients. The notation $f_i(\cdot)$ represents the implicit RMS dynamic response of the feeder rather than a closed-form static mapping. In particular, it corresponds to the solution of the coupled differential-algebraic system formed by the network algebraic equations, inverter inner-loop dynamics, and fault ride-through controls. The vector $\boldsymbol{\xi}(t)$ includes disturbance-related and operating-condition variables. This formulation captures the general objective of post-fault voltage recovery: minimizing voltage violations and voltage deviations after fault clearing while using feasible and limited reactive power support.

III. METHODOLOGY

The post-fault voltage recovery problem formulated in Section II is difficult to solve online using a conventional optimization-based controller, because the voltage response depends on the nonlinear dynamics of the distribution feeder, inverter capability limits, disturbance location, and operating condition. Moreover, the required control actions must be computed repeatedly during the recovery period after fault clearing. To address this challenge, this work adopts a MARL framework, where distributed PV inverters learn to coordinate their reactive power support based on local observations and a shared system-level performance objective.

A. Multi-Agent Control Setting

The controllable PV inverters $\mathcal{C}$ are modeled as agents. At each post-fault control step t , agent $k \in \mathcal{C}$ observes local electrical quantities and selects a reactive power action. The joint action changes the PV inverter reactive power injections and shapes the feeder voltage recovery trajectory.

The MARL problem is described by

$$(\mathcal{S}, \{\mathcal{O}_k\}_{k \in \mathcal{C}}, \{\mathcal{A}_k\}_{k \in \mathcal{C}}, P, r, \gamma), \quad (11)$$

where $\mathcal{S}$ is the global state space, $\mathcal{O}_k$ and $\mathcal{A}_k$ are the observation and action spaces of agent k , P denotes the RMS simulation dynamics, r is the reward, and γ is the discount factor.

During execution, each agent uses only local measurements. In this work, the observation of agent k is

$$o_k(t) = [v_k(t), p_k^{\text{PV}}(t), q_k(t-1), q_k^{\text{max}}(t)], \quad (12)$$

which includes local voltage, PV active power, previous reactive power, and available reactive power capability. Additional neighbouring-voltage or feeder-level measurements could be added through peer-to-peer communication or a feeder-level measurement concentrator, but this study focuses on the fully local observation setting.

B. Action Space and Inverter Constraints

Each agent controls the reactive power setpoint of its corresponding PV inverter through a normalized action

$$a_k(t) \in [-1, 1]. \quad (13)$$

The action is mapped to a feasible reactive power command using the inverter capability defined in Section II:

$$q_k(t) = a_k(t)q_k^{\text{max}}(t). \quad (14)$$

The joint action of all agents is

$$\mathbf{a}(t) = [a_1(t), a_2(t), \dots, a_{|\mathcal{C}|}(t)]^\top. \quad (15)$$

After applying the joint action, the RMS simulation advances to the next control step and returns the updated voltage response.

C. Reward Function

The reward encourages fast post-fault voltage recovery while limiting unnecessary reactive power usage. Using the voltage violation, voltage tracking error, and reactive power effort terms defined in Section II, the instantaneous reward is written as

$$r(t) = -(\lambda_{\text{vio}}J_{\text{vio}}(t) + \lambda_{\text{err}}J_{\text{err}}(t) + \lambda_qJ_q(t)), \quad (16)$$

where λ_{vio} , λ_{err} , and λ_q are weighting coefficients.

Two reward-shaping variants are used to study different control trade-offs. Fast MARL prioritizes rapid voltage recovery with stronger early reactive support, whereas Balanced RL applies higher penalties on reactive power effort, late positive reactive power injection, and post-fault oscillations. In the implementation, the compact terms in (16) are expanded into undervoltage, soft-overvoltage, hard-overvoltage, and time-dependent reactive-power penalties. The main values used for both policies are reported in Table II.

D. Training and Execution

The MARL controller is trained offline through repeated interaction with the RMS simulation environment. Each episode represents one post-fault recovery scenario of the IEEE 33-bus ADN and includes pre-fault, fault-on, and post-fault intervals. At the beginning of each episode, the operating condition, PV generation, load demand, and fault scenario are initialized.

During the fault-on interval, the supervisory MARL action is held fixed so that fast inverter inner controls, protection response, and fault ride-through dynamics respond without interference. After fault clearing, the MARL controller updates the reactive power setpoints of the controllable PV inverters at the supervisory control interval.

The policy of inverter agent k is denoted by

$$a_k(t) = \pi_k(o_k(t); \theta_k), \quad (17)$$

where $o_k(t)$ is the local observation, $a_k(t)$ is the normalized reactive power action, and θ_k denotes the policy parameters. During training, actions are sampled from a stochastic policy,

$$a_k(t) \sim \pi_k(\cdot | o_k(t); \theta_k), \quad (18)$$

then clipped to $[-1, 1]$ and mapped to the feasible inverter command

$$q_k(t) = a_k(t)q_k^{\text{max}}(t). \quad (19)$$

The training follows a centralized-training and decentralized-execution structure. During training, a shared system-level reward is computed from the feeder voltage response and reactive power usage. The RMS environment provides all bus voltages offline to compute $J_{\text{vio}}(t)$ and $J_{\text{err}}(t)$, while execution only requires each inverter agent to use its local observation for action selection. The training process is summarized in Algorithm 1.

The training objective is to maximize the expected cumulative discounted reward over the post-fault recovery horizon:

$$\max_{\{\theta_k\}_{k \in \mathcal{C}}} \mathbb{E} \left[\sum_{t \in \mathcal{T}_{\text{pf}}} \gamma^t r(t) \right]. \quad (20)$$

The proposed controller is empirical and does not provide formal closed-loop stability or voltage-security guarantees. PPO is used because its clipped surrogate objective restricts policy updates and improves training stability compared with unconstrained policy-gradient updates. After training, the learned policies are used deterministically, and no online optimization is required during deployment.

IV. NUMERICAL ANALYSIS

This section presents the numerical evaluation of the proposed MARL-based supervisory controller for post-fault voltage recovery. The objective is to assess whether coordinated inverter reactive power control can improve the dynamic voltage response after fault clearing compared with baseline operation and fixed-parameter control alternatives. The evaluation is performed using time-domain RMS simulations, where different operating conditions and fault scenarios are

Algorithm 1 PPO-Based MARL for Post-Fault Voltage Recovery

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1: Initialize actor-critic networks for all inverter agents
2: for each training iteration do
3:   Initialize rollout storage
4:   for each rollout episode do
5:     Initialize operating condition and fault scenario
6:     Run RMS simulation through pre-fault and fault-on
       periods, holding supervisory MARL actions fixed
7:     Clear the fault at time  $t_c$ 
8:     for each post-fault control step  $t \in \mathcal{T}_{pf}$  do
9:       Each agent observes  $o_k(t)$  and samples  $a_k(t)$  from
       the stochastic policy
10:      Clip  $a_k(t) \in [-1, 1]$  and map it to  $q_k(t) =$ 
        $a_k(t)q_k^{\max}(t)$ 
11:      Apply  $\mathbf{q}(t)$ , advance the RMS simulation, and
       obtain  $\mathbf{v}(t+1)$ 
12:      Compute shared reward  $r(t)$  and store the rollout
       transition
13:    end for
14:  end for
15:  Estimate returns and advantages using GAE with
        $\lambda_{GAE} = 0.97$ 
16:  Update actor-critic networks using the PPO clipped
       objective
17: end for
18: Return trained policies  $\{\pi_k(\cdot; \theta_k)\}_{k \in \mathcal{C}}$ 

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considered. The controllers are compared using post-fault voltage recovery indicators, including recovery time, settling time, post-fault voltage error, voltage violation area, time spent within the admissible voltage band, and reactive power effort.

A. Test System, Dataset, and Training Setup

The proposed control scheme is evaluated on a modified IEEE 33-bus distribution network, shown in Fig. 1. PV units are connected to selected buses and act as controllable inverter agents during the post-fault recovery interval. Load profiles are obtained from three years of Portuguese electricity consumption data [15], while PV generation profiles are taken from the Elia Group real-time solar dataset [16]. Both datasets are interpolated to a 3-minute resolution and mapped to the IEEE 33-bus feeder according to the nominal load and PV allocation of the buses. These data are used to sample realistic pre-fault operating points; within each short RMS episode, the sampled load and PV active power values are held constant, while the sub-second voltage dynamics are produced by the RMS network and inverter models.

For each training episode, an operating point is sampled from the time-series dataset and combined with a fault scenario. The simulation includes pre-fault, fault-on, and post-fault intervals. The supervisory MARL action is held fixed during the fault-on period and activated after fault clearing. The discount factor is set to $\gamma = 0.995$, corresponding to a

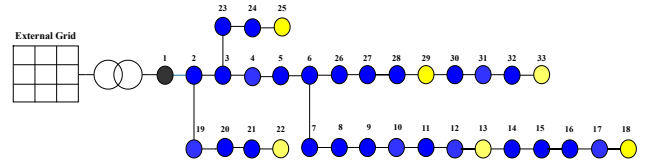


Fig. 1. Modified IEEE 33-bus ADN used for post-fault voltage recovery studies.

TABLE I
MAIN MARL TRAINING SETUP

Item	Setting
Agents	PV buses 12, 17, 21, 24, 28, 32
Training algorithm	PPO
Training iterations	120
Discount factor γ	0.995
GAE parameter λ_{GAE}	0.97
Actor/Critic learning rate	$10^{-4} / 4 \times 10^{-4}$
Voltage limits	[0.95, 1.05] pu

TABLE II
MAIN DIFFERENCES BETWEEN FAST MARL AND BALANCED MARL

Term	Fast MARL	Balanced MARL
Voltage tracking	1.0	1.2
Undervoltage penalty	8.5	7.0
Overvoltage penalty	7.0	8.0
Voltage-change penalty	1.0	1.4
$ Q $ penalty range	0.02–0.35	0.05–0.75
Q ramp penalty	0.08	0.12

near-undiscounted objective over the short recovery horizon. The main training settings are summarized in Table I.

The compact reward in (16) is implemented with separate voltage-violation, voltage-change, and time-dependent reactive-power penalties. Table II summarizes the main differences between the two policies. Fast MARL uses lower reactive-power penalties to prioritize recovery speed, whereas Balanced MARL applies stronger effort and voltage-change penalties to reduce peaks and oscillations.

B. Results

The proposed post-fault control strategies are first analyzed using a representative RMS simulation scenario to illustrate their dynamic behavior. The evaluation is then extended to 101 RMS simulation scenarios, consisting of one nominal case and 100 randomized operating conditions and fault realizations. Four controllers are compared: droop control, fixed PI, Fast MARL, and Balanced MARL.

The droop controller follows an inverter-based transient voltage-support structure, where reactive power is adjusted according to local voltage deviation [17]. It uses $v^{\text{ref}} = 1.0$ pu, droop gain $k_q = 3.0$, and a reactive power command limit of 0.60 pu. The fixed-PI controller is implemented as a non-adaptive PCC-voltage supervisor. Its input is $e_{PCC}(t) = v^{\text{ref}} - v_{PCC}(t)$, with $v^{\text{ref}} = 1.0$ pu and a deadband of 0.0015 pu. Fixed gains $K_p = 2.2$ and $K_i = 8.0$ are used for all scenarios, with output saturation at ± 0.60 pu, anti-windup correction, a 0.04 s output filter, and a 4.0 pu/s output rate limit. During the fault-on interval, the command is held fixed, so the controller

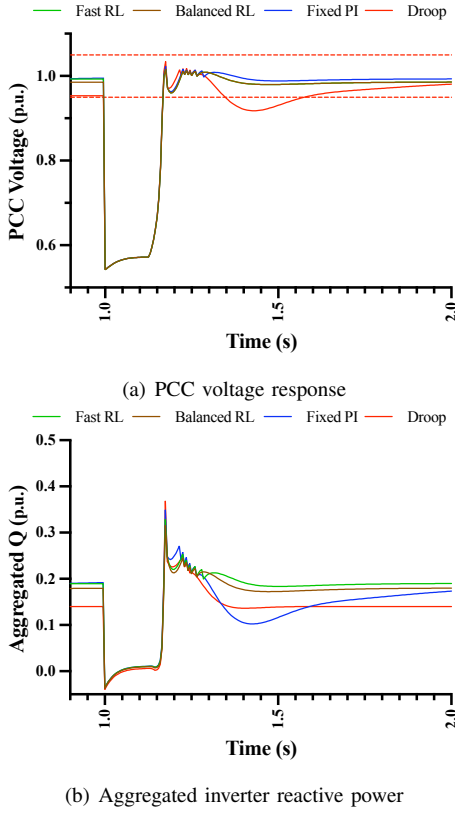


Fig. 2. Representative post-fault response under droop, fixed PI, and MARL-based supervisory control.

remains a supervisory post-fault regulator and does not alter inverter inner-loop or fault ride-through dynamics [18]. The two MARL policies correspond to different reward-shaping choices: Fast RL prioritizes rapid post-fault recovery, whereas Balanced RL applies stronger penalties on reactive power effort and late post-fault oscillations.

Fig. 2 shows the PCC voltage and aggregated inverter reactive power response for the representative post-fault scenario. The droop controller exhibits a larger voltage excursion after fault clearing and a slower return toward the nominal operating region. The fixed PI controller improves the final voltage value, but its response contains a more persistent transient. In contrast, the MARL-based controllers provide smoother post-

fault voltage recovery. Fast MARL rapidly drives the voltage back toward the admissible region and achieves the lowest post-fault mean voltage error in this case, while Balanced MARL produces a more conservative response with lower voltage peak and lower maximum reactive power injection.

The representative metrics in Table III confirm the visual trends in Fig. 2. Fast MARL obtains the lowest post-fault mean voltage error, 0.009 pu, and the highest time-in-band fraction, 0.997. Balanced MARL gives the lowest maximum PCC voltage, 1.016 pu, and the lowest maximum reactive power injection, 0.316 pu. This illustrates the intended distinction between the two learned policies: Fast MARL prioritizes recovery quality, whereas Balanced MARL limits voltage peaks and reactive power usage. Although the droop controller has the lowest post-fault $|Q|$ area, this mainly reflects weaker corrective action rather than better recovery.

TABLE III
KEY METRICS FOR THE REPRESENTATIVE POST-FAULT SCENARIO

Metric	Droop	PI	Fast	Bal.
Max. PCC V [pu] ↓	1.051	1.034	1.023	1.016
Final $ V - 1 $ [pu] ↓	0.047	0.006	0.006	0.014
Time in-band ↑	0.994	0.869	0.997	0.997
Mean $ V - 1 $ [pu] ↓	0.045	0.024	0.009	0.016
Max. Q [pu] ↓	0.369	0.349	0.329	0.316
Post-fault $ Q $ area ↓	0.263	0.313	0.347	0.329

Table IV summarizes the 101-scenario evaluation using mean and standard-deviation values and reports two aggregate indicators computed over the same ten metrics. The aggregate ranking score assigns 4 to 1 points for each metric, with 4 assigned to the best-performing controller according to the desired optimization direction. Since this score only uses controller ordering and not the magnitude of the differences, a normalized aggregate score is also reported by min-max normalizing controller means for each metric before summation. Both aggregate indicators are used only as compact summaries in the absence of operator-specific metric weights; the individual metrics remain the primary basis for interpretation.

Overall, the droop controller gives the weakest recovery performance. It has the lowest final in-band success rate, the highest final voltage error, and the lowest post-fault time-in-band fraction. This indicates that conventional local droop action is often insufficient to guarantee satisfactory post-fault

TABLE IV
MEAN PERFORMANCE OVER 101 EVALUATION SCENARIOS

Metric	Droop	FixedPI	Fast MARL	Balanced MARL
Final in-band flag ↑	0.4356 ± 0.4983	1.0000 ± 0.0000	0.9604 ± 0.1960	0.9015 ± 0.3574
Final voltage abs. error [pu] ↓	0.0513 ± 0.0047	0.0058 ± 0.0009	0.0158 ± 0.0047	0.0325 ± 0.0150
First recovery time [s] ↓	0.0628 ± 0.0160	0.0453 ± 0.0085	0.0520 ± 0.0065	0.0752 ± 0.0159
Settling time [s] ↓	0.2015 ± 0.1565	0.4203 ± 0.1240	0.0712 ± 0.0162	0.0730 ± 0.0154
Post-fault time in-band fraction ↑	0.4576 ± 0.4299	0.8620 ± 0.0452	0.9391 ± 0.1447	0.8539 ± 0.2741
Post-fault mean abs. voltage error [pu] ↓	0.0623 ± 0.0050	0.0366 ± 0.0050	0.0312 ± 0.0070	0.0446 ± 0.0134
Post-fault voltage violation area ↓	0.0291 ± 0.0065	0.0285 ± 0.0049	0.0252 ± 0.0039	0.0262 ± 0.0051
Maximum PCC voltage [pu] ↓	1.0184 ± 0.0171	1.0212 ± 0.0078	1.0097 ± 0.0107	1.0061 ± 0.0114
Maximum Q injection [pu] ↓	0.3974 ± 0.0335	0.4032 ± 0.0390	0.3634 ± 0.0391	0.3541 ± 0.0390
Post-fault $ Q $ area ↓	0.2937 ± 0.0259	0.3669 ± 0.0404	0.3585 ± 0.0083	0.3402 ± 0.0073
Aggregate ranking score ↑	18.0	25.0	31.0	26.0
Normalized aggregate score ↑	2.3432	5.0000	7.2177	6.7696

recovery across varying operating and disturbance conditions.

The fixed-PI controller achieves the best final voltage accuracy, the shortest first recovery time, and a final in-band flag of 1.0000. However, it also gives the longest settling time, indicating a prolonged post-fault transient. The FixedPI baseline should therefore be interpreted as a fixed-parameter supervisory comparator rather than an optimal or grid-code-certified controller. In contrast, the MARL policies are evaluated with the same trained parameters across all scenarios and provide more favorable overall dynamic recovery behavior without scenario-specific retuning.

The Fast MARL policy provides the strongest overall recovery-oriented behavior. It achieves the highest post-fault time-in-band fraction, the lowest post-fault voltage violation area, and the shortest settling time. This is also reflected in both aggregate indicators, where Fast RL obtains the highest aggregate ranking score and normalized aggregate score. The Balanced MARL policy provides a more conservative response, yielding the lowest maximum PCC voltage and maximum reactive power injection, while also achieving the second-best normalized aggregate score.

These results show that the MARL gain is mainly concentrated on settling behavior, post-fault in-band operation, violation-area reduction, and peak-control moderation, rather than final voltage accuracy or first recovery time. Thus, individual metrics should not be interpreted in isolation. Overall, Fast MARL offers the strongest recovery-oriented behavior, while Balanced MARL provides a more conservative trade-off with lower voltage peaks and reduced reactive power injection.

V. CONCLUSION

This paper proposed a Multi-Agent Reinforcement Learning (MARL)-based supervisory controller for post-fault voltage recovery in Active Distribution Networks (ADNs) with PV inverter resources. The controller coordinates inverter reactive power setpoints after fault clearing while keeping the supervisory action fixed during the fault-on interval to avoid interference with fast inverter and fault ride-through dynamics. It was evaluated through RMS simulations on a modified IEEE 33-bus feeder over 101 operating and fault scenarios and compared against droop and fixed-PI baselines.

The results show that the MARL policies improve post-fault recovery mainly in terms of settling behavior, post-fault in-band operation, voltage violation area, and voltage peak reduction. Fast MARL achieved the best overall recovery-oriented performance, while Balanced MARL provided a more conservative response with lower voltage peaks and reduced peak reactive power injection. The fixed-PI controller achieved strong final voltage accuracy, but with significantly longer settling time. The present formulation handles voltage limits through reward penalties and therefore does not provide hard safety guarantees. Future work will extend the framework with constrained or safe MARL methods, centralized MPC or post-fault optimal reactive power control benchmarks, topology-aware observations, feeder reconfiguration studies, additional fault cases, and validation on larger ADNs.

ACKNOWLEDGMENT

This project has received funding from the European Union's Horizon Europe Framework Programme (HORIZON) under the GA n. 101120278 - DENSE.

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